

Quiz (Habanero)

- Q1.** If x represents temperature in C and y represents rainfall in mm, which system of equations models the relationship between temperature and rainfall in a region?
- (A) $x + y = 20$, $y = 2x - 10$
 (B) $x - y = 15$, $y = x + 5$
 (C) $2x + y = 25$, $y = x + 10$
 (D) $x - 2y = 8$, $y = 3x - 15$
 (E) $x + 2y = 30$, $y = 2x + 5$
- Q2.** What is the intersection point of the lines $y = 2x - 3$ and $y = x + 2$?
- (A) (1, -1)
 (B) (2, 1)
 (C) (5, 7)
 (D) (-1, -5)
 (E) (3, 3)
- Q3.** Solve the system: $x^2 + y^2 = 20$, $y = x - 2$. What is x ?
- (A) -2, 4
 (B) 1, 3
 (C) -1, 5
 (D) 2, 6
 (E) 0, 4
- Q4.** The ecosystem of a forest is modeled by $y = x^2 - 4x + 3$ and $y = 2x - 5$. What is the y -coordinate of the intersection point?
- (A) -1
 (B) 1
 (C) 3
 (D) 5
 (E) 7
- Q5.** Solve $y = 3x - 2$ and $y = x^2 + x - 2$ for x . Which statement is true?
- (A) One solution
 (B) Two distinct solutions
 (C) No real solutions
 (D) Infinite solutions
 (E) Two repeated solutions
- Q6.** A quadratic equation models the relationship between temperature (x) and rainfall (y) in a region: $y = x^2 - 3x + 2$. A linear equation is $y = x + 1$. What is the x -coordinate of their intersection?
- (A) 1
 (B) 2
 (C) 3
 (D) -1
 (E) 0
- Q7.** Temperature changes (x) affect rainfall (y): $x^2 - 2x - 3 = y$, $2x - y = 5$. What is the value of y when the system is solved?
- (A) -1
 (B) 1
 (C) 3
 (D) -3
 (E) 5
- Q8.** The quadratic equation $y = x^2 - x - 1$ and the linear equation $y = 2x + 3$ intersect at what point?
- (A) (1, 2), (-2, -1)
 (B) (-1, 3), (3, 9)
 (C) (2, 5), (-1, 1)
 (D) (-2, 1), (3, 7)
 (E) (0, 3), (2, 7)

Open Ended Questions (FRQ)

- Q9.** Solve the system of equations: $y = x^2 + 2x - 1$, $y = 3x - 2$. Show all steps and give final answers for x and y . Provide intersection points if they exist.
- Q10.** The temperature (x) and rainfall (y) of a region are modeled by two equations: $y = x^2 - 4x + 5$, $y = x + 1$. Find the x - and y -coordinates of the points of intersection, and interpret the results in the context of the problem.

Chef's Tips

- Emphasize solving systems of linear and quadratic equations to find intersection points.
- Discuss how these systems apply to weather and environmental models.
- Provide examples that demonstrate unique solutions and repeated or no solutions.